Conditional Gaussian Enhanced Dense Correlation Matching for Cross-Category Land Cover Classification

Huan Ni, Haiyan Guan, *Senior Member, IEEE*, Xudong Tong, and Jocelyn Chanussot, *Fellow, IEEE*

***Abstract*— As the requirements for the downstream tasks of land cover classification (LCC) continue to increase, the category system used for LCC is constantly being refined. This causes previous land cover products and manually annotated training samples to become quickly outdated. Meanwhile, manually annotating samples with more refined categories is extremely time-consuming. To address this impasse, a cross-category knowledge transfer process is needed that can directly generate land cover products under a fine-scale category system using existing training samples under a large-scale category system. Accordingly, this article proposes a cross-category LCC method called conditional Gaussian enhanced dense correlation matching (CGE-DCM). CGE-DCM uses samples under a large-scale category system for training. It then uses only one annotated example of each fine-scale category to achieve fine-scale LCC. In cases with very little sample support, the problems caused by different spectra of the same object and different objects of the same spectrum in complex scenes can be particularly severe. To solve this problem and improve the accuracy of classifications of complex objects, CGE-DCM offers a dense correlation matching (DCM) strategy. In addition, context distribution is different under fine-scale category systems than it is under large-scale category systems. For this reason, CGE-DCM features a conditional Gaussian enhancement mechanism and designs different loss functions for degenerated and nondegenerated scenarios to ensure the stability of the model. Extensive experiments on Gaofen-2 and Orbita hyperspectral satellite (OHS) images demonstrate the effectiveness of each module in CGE-DCM and its superiority over existing methods.**

***Index Terms*— Conditional Gaussian, cross-category, land cover classification (LCC), matching.**

I. Introduction

REMOTE sensing-based land cover classification (LCC) has received much attention because of its wide-ranging applications [1]. LCC assigns a land cover category to each pixel in a remote sensing image, which helps us better understand the relationship between humans and nature [2]. On the one hand, from the perspective of data sources, the literature on LCC has variously focused on the mapping of multitemporal [3], multimodal [4], and single-form data [5]. On the other hand, from a methodological perspective, the extant research can be divided into fully supervised [6], weakly supervised [5], [7], and unsupervised domain adaptation [8] classifications. These studies have achieved remarkable results in improving the intelligence of LCC methods by enhancing their generalization ability when there are insufficient samples or differences in resolutions, data sources, or geographic locations. However, we have found that, with the development of the geographic information industry, the categories of LCC products are required to be more and more refined. Therefore, some high-resolution LCC datasets, such as the gaofen image dataset (GID), provide hierarchical category systems: the first-level system contains only large-scale categories, and the second-level system contains fine-scale categories [9]. Table I presents the hierarchical category systems of GID. Here, the finer the classification, the higher the cost of manual labeling. In this case, it is important to understand how categories can be updated to form a more refined system if the land cover samples contain only large-scale categories. This article defines this task as the cross-category LCC problem.

Cross-category LCC is difficult because it involves training on samples of large-scale categories and then equipping the model with the ability to recognize fine-scale categories. In this case, only the large-scale categories are known by the model, and the fine-scale categories are unknown. Here, we allow for only one annotated example of each fine-scale category in the model when classifying land cover under the fine-scale category system. This process is similar to the few-shot semantic segmentation process used for unknown categories [10], [11]. Few-shot semantic segmentation involves generalizing the segmentation's capability to unknown categories with the support of only a few annotated data [12], [13]. Prototype methods that divide the data into support and query sets and match support prototypes with query features to make dense predictions are popular in few-shot segmentation [14]. These prototype methods have been applied in the field of remote sensing in, for instance, target recognition [15], [16], hyperspectral image analysis [17], and semantic segmentation [18], [19]. However, although these studies are similar to those involving cross-category LCC tasks, they have the following differences.

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TABLE I
HIERARCHICAL CATEGORY SYSTEMS OF GID

Large-scale categories	Fine-scale categories
Built-up	Industrial land Urban residential Rural residential Traffic land
Farmland	Paddy field Irrigated land Dry cropland Garden plot
Forest	Arbor woodland Shrub land
Meadow	Natural grassland Artificial grassland
Water	River Lake Pond

1) The target of the recognition task is only considered for a limited number of specific categories, and label information is assigned to only a small number of pixels in the image, as can be seen in the results presented in [20].
2) The category system used by studies of few-shot semantic segmentation is different than that used in LCC tasks.
3) The mean intersection over union (mIoU) of a few-shot segmentation is computed based on the binary classification results for each category, without accounting for the confusion that can occur with multiple categories; this is not the case with the mIoU in LCC. Therefore, existing studies do not provide an adequate basis for evaluating the performance of few-shot segmentation in LCC tasks.

More importantly, the current literature faces the following two dilemmas: On the one hand, considerable intraclass diversity is challenging for the few-shot segmentation approach due to the lack of examples of unknown categories. To solve this problem, Yang et al. [21] built prototype mixture models to enhance the ability of the prototype to represent complex targets based on feature-clustering results. Yet how can the number of clusters be determined when we do not know the complex form of the target? On the other hand, because there are no unknown categories in the training samples, the unfamiliar context structures of these categories make it difficult for the support prototype of the unknown category to be sufficiently generalized in the prediction process. To address this issue, Fan et al. [22] designed a self-support matching strategy to effectively capture the consistent underlying characteristics of query objects and thus suitably match query features. Cheng et al. [11] proposed a cross-referenced decoder to reweigh features and aggregate multilevel features to meet the same objective. Although these studies have made positive contributions, they do not provide an explainable solution to this issue.

We present an interpretable mechanism for addressing these issues. To tackle the first problem, a dense correlation matching (DCM) strategy for cross-category LCC is proposed. In DCM, feature-correlation matrices exploit classification clues. Based on these clues, the matching process can make a dense comparison of correlations between support and query features to obtain more accurate classification results. To surmount the second problem, we introduce conditional Gaussian distribution to simulate prototype properties in the presence of unknown contexts, yielding conditional Gaussian-based enhancement (CGE). To eliminate the instability caused by unknown feature distribution, CGE adds a regularization term that constrains the feature distribution through maximum likelihood estimation. The main contributions of our work can be summarized as follows.

1) DCM fully exploits classification clues based on feature-correlation matrix generation (FcmG) and dense matching (DenseM). By densely comparing the feature-correlation matrices of the support and query features using the Frobenius norm, more accurate cross-category LCC results are obtained.
2) CGE introduces conditional Gaussian properties to enhance prototypes in the presence of unknown contexts. Furthermore, CGE constructs a regularization term that constrains the feature distribution through maximum likelihood estimation.
3) A cross-category LCC method called conditional Gaussian enhanced DCM (CGE-DCM) is proposed. The performance of CGE-DCM on unknown categories in a cross-category LCC task is validated.

II. RELATED WORK

Two areas of research are relevant to this article: LCC and few-shot segmentation.

A. Land Cover Classification

LCC aims to associate each pixel in a remote sensing image with a predefined land cover category [9]. A range of machine learning methods, such as random forests [23], hidden Markov model [24], Gaussian process [25], semisupervised predictive clustering trees [26], knowledge distillation [27], and deep learning models [8], [28] have been modified to meet this objective. Among these methods, deep learning models have been shown to be superior to traditional machine learning classifiers in LCC tasks [29], [30]. Advanced network architectures such as [31] have been introduced. Currently, LCC has been developed to have a higher resolution [32] and stronger generalization ability [33].

To pursue a higher resolution, there are approaches that can be taken. First, Tong et al. [32] and Xie et al. [34] directly used high-resolution remote sensing images and designed a deep learning network that was efficient for detailed information extraction and for producing high-resolution land cover products. Second, as shown in the studies of Li et al. [7] and Chen et al. [35], here based on low-resolution outdated land cover products, a cross-resolution deep learning network was designed to update land cover products to improve the resolution. For stronger generalization ability, transferable deep learning models have been introduced [9]. Transferable deep learning models usually introduce domain adaptation methods

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as a way to eliminate the data disturbance issue caused by different image acquisition conditions and different geographical areas [33], [36]. These domain adaptation mechanisms are usually closely related to remote sensing data types, such as S2FL [37] and SSFR [38] for hyperspectral remote sensing images and DAGDNet [1] and LCGDM [33] for optical remote sensing images. Among these methods, LCGDM [33] is representative. LCGDM first proved that local consistency and global diversity could be modeled using the Frobenius norm and nuclear norm, and LCGDM also introduced a new measure of local consistency and global diversity that was integrated into the domain adaptation method. Although studies, such as LCGDM [33] and ClA-UDA [39], have improved the LCC outcomes, they are still unable to update the category system of land cover products, and the rapid updates of the category system usually require few-shot segmentation.

B. Few-Shot Segmentation

Shaban et al. [40] laid the foundation for few-shot segmentation methods. Specifically, Shaban et al. developed a data structure that consisted of support and query images and defined therein the one-shot segmentation problem. Dong and Xing [41] inherited and developed the one-shot segmentation framework [40], proposing the prototype method, which provides the basis for the widely used few-shot segmentation framework. This framework first uses CNNs to generate a prototype of each category based on a support image. To segment the query image, the similarity between each feature vector of the query image and the prototype is calculated [12], [42]. Zhang et al. [43] proposed a masked average pooling method for prototype extraction, which can more successfully extract foreground and background information from a support set. Furthermore, they introduced cosine similarity to measure the distance between prototypes and query features. The cosine similarity has been widely adopted in later studies. Their study was officially published in 2020, but its preprint version was made public in 2018 and received widespread attention. Later, scholars determined that this framework does not distinguish between the processes of knowledge extraction and segmentation. In response, PANet [44] and CRNet [45] were proposed, which successfully separated the process of knowledge extraction and segmentation, and which modeled the relationship between support and query images more systematically. Subsequently, studies were carried out from the perspectives of multiresolution [46], rotation-invariant properties [47], frequency prototype generation [48], and self-attention mechanisms [49].

A series of recent studies, such as those that introduced SAML [50], CANet [51], and PFENet [52], further found that the problem of intraclass diversity hinders the improvement of prototype representation ability. These methods have achieved some success in solving this problem. However, the problem of semantic ambiguity stemming from global average pooling in the process of extracting prototypes remains unresolved. Therefore, Liu et al. [53], Yang et al. [21], and Yao et al. [54] proposed multiprototype models, modeling the correlation between prototypes in different ways. As we asked in Section I, it is difficult to determine the number of prototypes when we do not know the complex form of the target. Furthermore, because the unknown categories do not exist in the training samples, the unfamiliar context structures of the unknown categories make it difficult for the support prototype of the unknown category to achieve sufficient generalization ability in the prediction process. To address this problem, self-supporting strategies [22], [55], self-regularizing prototypes [14], a distillation mechanism [16], and a feature reweighting process [11] have been proposed. The methods of Fan et al. [22] and Cheng et al. [11], as shown in Section I, are representative. Notably, the method described by Li et al. [56] not only addressed the intraclass diversity issue, but also helped tackle the issue of the unfamiliar context structures of the unknown categories. However, the method did not provide an explainable solution for this issue.

III. Methods

A. Problem Setting

In the cross-category LCC task, the training samples only contain the large-scale land cover categories. After the model has been trained, only one annotated example is provided for each fine-scale category to map the fine-scale land cover. This process is similar to a special case of few-shot segmentation, namely, one-shot segmentation, which is defined in [40]. One-shot segmentation uses the episode paradigm proposed in [57]. Each episode $\mathcal{E}$ consists of a support set $\mathcal{S}$ and a query set $\mathcal{Q}$, that is, $\mathcal{E} = \{\mathcal{S}, \mathcal{Q}\}$. In the one-shot setup, both $\mathcal{S} = \{I^s, Y^s\}$ and $\mathcal{Q} = \{I^q, Y^q\}$ are composed of an image-label pair. I^s is the support image, and Y^s is the ground-truth of I^s. I^q is the query image, and Y^q is the ground-truth of I^q. As the model is training, $\mathcal{S}$ and $\mathcal{Q}$ in each episode are randomly selected from samples with large-scale categories. When the model makes a prediction, only one instance for each fine-scale category is annotated in a fixed support image I^s, and thus, $\mathcal{S} = \{I^s, Y^s\}$ can be constructed. With this minimal supporting information, given a new image, the model can achieve LCC in the fine-scale category system.

B. Methods Overview

The overview of the CGE-DCM for cross-category LCC is shown in Fig. 1. The DCM and CGE modules are the key components of CGE-DCM. In terms of the training process, given an episode $\mathcal{E} = \{\mathcal{S}, \mathcal{Q}\}$ that is randomly selected from the large-scale categories, where $\mathcal{S} = \{I^s, Y^s\}$ and $\mathcal{Q} = \{I^q, Y^q\}$, a feature generator (such as one of the ResNet series [58]) is performed to extract features. As a result, the support feature f^s for I^s and the query feature f^q for I^q are generated. f^s and f^q are put into DCM, in which the feature-correlation matrix is computed and densely compared. In each training iteration, DCM produces a prediction P_k^{DCM} of I^q for a category that is labeled as k. Then, CGE is performed based on the results of DCM. CGE enhances the prototypes based on the conditional Gaussian and produces the enhanced prototypes $\hat{\xi}_k^f$ for the foreground and $\hat{\xi}_k^b$ for the background. Furthermore, CGE provides a regularization term $\mathcal{L}_k^R$ for back-propagation to normalize the data distributions.

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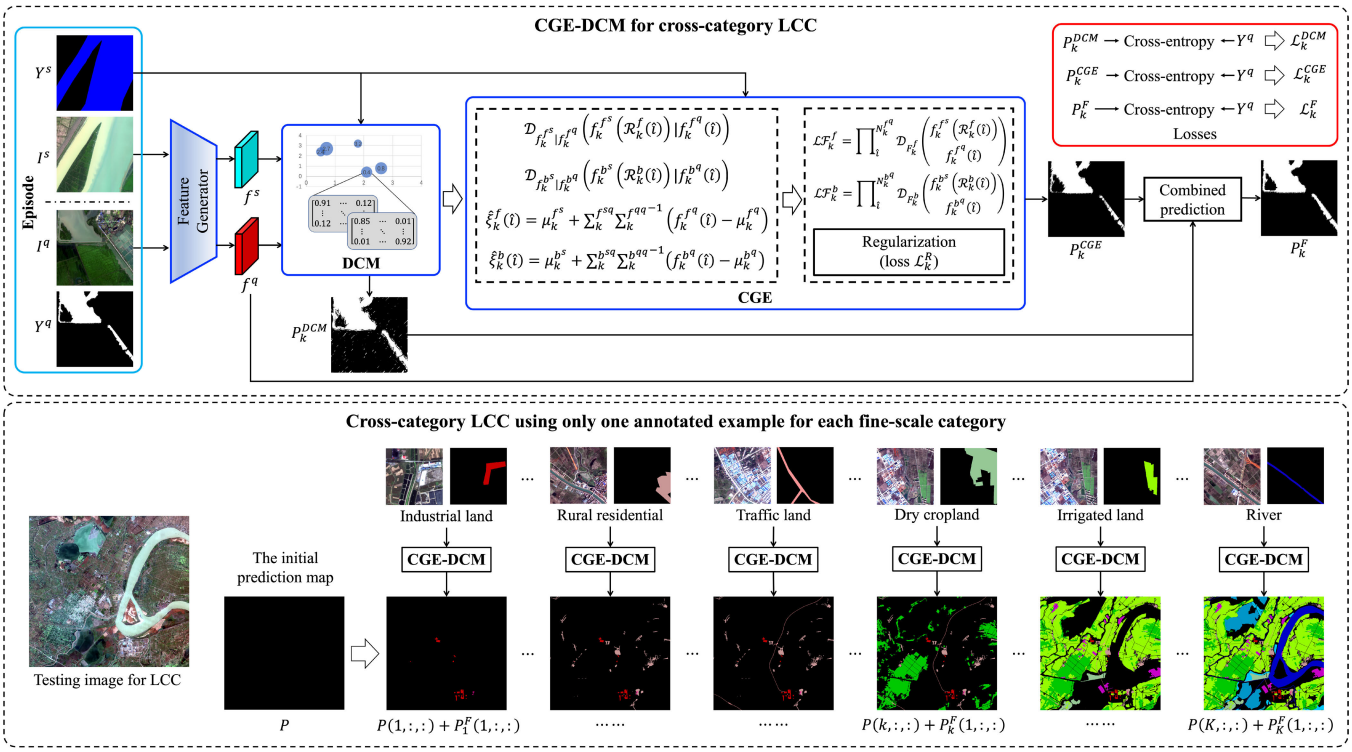


Fig. 1. Overview of the CGE-DCM for cross-category LCC.

Based on $\hat{\xi}_k^f$ and $\hat{\xi}_k^b$, a prediction P_k^{CGE} of I^q is generated. Finally, f^q, P_k^{DCM}, and P_k^{CGE} are put into the combined prediction layer to produce the final prediction P_k^F of I^q. The loss for back-propagation consists of four terms: $\mathcal{L}_k^R$, $\mathcal{L}_k^{\mathrm{DCM}}$, $\mathcal{L}_k^{\mathrm{CGE}}$, and $\mathcal{L}_k^F$. $\mathcal{L}_k^{\mathrm{DCM}}$, $\mathcal{L}_k^{\mathrm{CGE}}$, and $\mathcal{L}_k^F$ are computed using the predictions P_k^{DCM}, P_k^{CGE}, and P_k^F. $\mathcal{L}_k^R$ is the regularization loss for CGE. In terms of the testing process, the flowchart of CGE-DCM is similar to that of the training process. However, there are two differences. First, the query image is the testing image, which does not contain the ground-truth. Second, the losses are ignored in the testing process. Given only one annotated example of each fine-scale category, the pixels in all the testing images belonging to the fine-scale category are recognized. Taking the recognition of rural residential in Fig. 1 as an example, the annotated example of rural residential is an instance which is annotated in a fixed image.

C. Dense Correlation Matching

Taking f^s, Y^s, and f^q as the inputs, DCM first divides f^s into foreground features $f_k^{f^s} \in \mathbb{R}^{N_k^{f^s} \times C^{sg}}$ and background features $f_k^{b^s} \in \mathbb{R}^{N_k^{b^s} \times C^{sg}}$ using the mask of the current kth category as determined by Y^s. It is worth noting that Y^s contains more than one category. This article processes each category in Y^s in an iterative manner. Here, we take the kth category as an example. Suppose $f^s(i)$ is the ith feature vector in f^s and $Y^s(i)$ is the ith element in Y^s; then, the division process is performed as follows:

$$\begin{cases} \text{Add } f^s(i) \text{ to } f_k^{f^s}, & \text{if } Y^s(i) = k \\ \text{Add } f^s(i) \text{ to } f_k^{b^s}, & \text{otherwise.} \end{cases} \tag{1}$$

Then, a linear layer is performed so that $f_k^{f^s} \in \mathbb{R}^{N_k^{f^s} \times C^{fc}}$, $f_k^{b^s} \in \mathbb{R}^{N_k^{b^s} \times C^{fc}}$, and $f^q \in \mathbb{R}^{N^q \times C^{fc}}$. Each feature vector in $f_k^{f^s}$, $f_k^{b^s}$, and f^q is normalized

$$\begin{cases} f_k^{f^s}(\bar{i}) = \dfrac{f_k^{f^s}(\bar{i}) - \mu^s}{\mathrm{Norm}\left(f_k^{f^s}(\bar{i})\right)}, & \bar{i} = 1, \ldots, N_k^{f^s} \\ f_k^{b^s}(\tilde{i}) = \dfrac{f_k^{b^s}(\tilde{i}) - \mu^s}{\mathrm{Norm}\left(f_k^{b^s}(\tilde{i})\right)}, & \tilde{i} = 1, \ldots, N_k^{b^s} \\ f^q(\hat{i}) = \dfrac{f^q(\hat{i}) - \mu^q}{\mathrm{Norm}\left(f^q(\hat{i})\right)}, & \hat{i} = 1, \ldots, N^q \end{cases} \tag{2}$$

where $\mathrm{Norm}(\cdot)$ is the ℓ_2 norm and

$$\begin{cases} \mu^s = \dfrac{\sum_{\bar{i}=1}^{N_k^{f^s}} f_k^{f^s}(\bar{i}) + \sum_{\tilde{i}=1}^{N_k^{b^s}} f_k^{b^s}(\tilde{i})}{N_k^{f^s} + N_k^{b^s}} \\ \mu^q = \dfrac{\sum_{\hat{i}=1}^{N^q} f^q(\hat{i})}{N^q}. \end{cases} \tag{3}$$

Subsequently, the correlation function is applied to $f_k^{f^s}(\bar{i})$, $f_k^{b^s}(\tilde{i})$, and $f^q(\hat{i})$, and a set of feature-correlation matrices are generated as follows:

$$\begin{cases} M_k^{f^s}(\bar{i}) = f_k^{f^s}(\bar{i}) f_k^{f^s}(\bar{i})^T \in \mathbb{R}^{C^{fc} \times C^{fc}}, & \bar{i} = 1, \ldots, N_k^{f^s} \\ M_k^{b^s}(\tilde{i}) = f_k^{b^s}(\tilde{i}) f_k^{b^s}(\tilde{i})^T \in \mathbb{R}^{C^{fc} \times C^{fc}}, & \tilde{i} = 1, \ldots, N_k^{b^s} \\ M^q(\hat{i}) = f^q(\hat{i}) f^q(\hat{i})^T \in \mathbb{R}^{C^{fc} \times C^{fc}}, & \hat{i} = 1, \ldots, N^q. \end{cases} \tag{4}$$

$M_k^{f^s}(\bar{i})$, $M_k^{b^s}(\tilde{i})$, and $M^q(\hat{i})$ are used as the basis for measuring similarity. To compute the similarities between $M_k^{f^s}(\bar{i})$ and

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$M^q(\hat{i})$ and between $M_k^{b^s}(\tilde{i})$ and $M^q(\hat{i})$, the Frobenius norm $\|\cdot\|_F$ is utilized

$$\begin{cases} S_k^{f^s}(\hat{i},\bar{i}) = \exp\left(-\|M_k^{f^s}(\bar{i}) - M^q(\hat{i})\|_F\right) \\ S_k^{b^s}(\hat{i},\tilde{i}) = \exp\left(-\|M_k^{b^s}(\tilde{i}) - M^q(\hat{i})\|_F\right). \end{cases} \tag{5}$$

For the $\hat{i}$th position in f^q, $S_k^{f^s}(\hat{i},\bar{i})$, $\bar{i} = 1,\ldots,N_k^{f^s}$ and $S_k^{b^s}(\hat{i},\tilde{i})$, $\tilde{i} = 1,\ldots,N_k^{b^s}$ are computed. We take the largest one-tenth of all the elements of $S_k^{f^s}$ and $S_k^{b^s}$ to form two similarity sets $\widehat{S_k^{f^s}}$ and $\widehat{S_k^{b^s}}$ corresponding to the foreground and background. The prediction of the $\hat{i}$th position is as follows:

$$\begin{cases} P_k^{\mathrm{DCM}}[0](\hat{i}) = \mathrm{Mean}\left(\widehat{S_k^{b^s}}\right) \\ P_k^{\mathrm{DCM}}[1](\hat{i}) = \mathrm{Mean}\left(\widehat{S_k^{f^s}}\right) \end{cases} \tag{6}$$

where $\mathrm{Mean}(\cdot)$ computes the mean value. If $P_k^{\mathrm{DCM}}[1](\hat{i})$ is larger than $P_k^{\mathrm{DCM}}[0](\hat{i})$, the $\hat{i}$th position is segmented into the kth category. After all the positions in f^q are processed as (2)–(6), the image-like feature map f^q is segmented, and the prediction P_k^{DCM} of DCM is obtained.

When determining $S_k^{f^s}$ and $S_k^{b^s}$, we also record the indices of the support feature vectors in $f_k^{f^s}$ and $f_k^{b^s}$ that are closest to each query feature vector in f^q, hence forming two best matching relationships $\mathcal{R}_k^f$ and $\mathcal{R}_k^b$, which are used to construct the subsequent joint probability distribution. In other words, given the $\hat{i}$th feature vector in f^q, the closest feature vector in $f_k^{f^s}$ is on the $\mathcal{R}_k^f(\hat{i})$th position, while the one in $f_k^{b^s}$ is on the $\mathcal{R}_k^b(\hat{i})$th position.

D. Conditional Gaussian-Based Enhancement

1) Conditional Enhancement: Given P_k^{DCM}, f^q can be divided into foreground features $f_k^{f^q}$ and background features $f_k^{b^q}$. Concretely,

$$\begin{cases} \text{Add } f^q(\hat{i}) \text{ to } f_k^{f^q}, & \text{if } P^{\mathrm{DCM}}[1](\hat{i}) > P^{\mathrm{DCM}}[0](\hat{i}) \\ \text{Add } f^q(\hat{i}) \text{ to } f_k^{b^q}, & \text{otherwise} \end{cases} \tag{7}$$

where $f^q(\hat{i})$ is the $\hat{i}$th feature vector of f^q. At the same time, $\mathcal{R}_k^f$ and $\mathcal{R}_k^b$ are also simplified. The indices corresponding to the query feature vectors that are added into $f_k^{f^q}$ are removed from $\mathcal{R}_k^b$, and the indices corresponding to the query feature vectors that are added into $f_k^{b^q}$ are removed from $\mathcal{R}_k^f$.

For large-scale categories, DCM enhances the stability of one-shot segmentation by fully considering intraclass diversity. However, when applied to unfamiliar contexts of fine-scale categories, its stability is still not guaranteed. We need to augment DCM with the detected features of unknown categories. Only in this way can we fully take into account the characteristics of currently unknown categories.

Suppose a set of random vectors X_1 has the mean vector μ_1 and the covariance matrix Σ_{11} and suppose a set of random vectors X_2 has the mean vector μ_2 and covariance matrix Σ_{22}. If $X = \binom{X_1}{X_2}$ with a covariance matrix $\begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$ follows a joint Gaussian distribution, the conditional Gaussian probability density of X_1 given X_2 is as follows [59]:

$$\mathcal{D}_{X_1|X_2}(x_1|x_2) = \frac{1}{(2\pi)^{n_1/2}\left(\det\left(\tilde{\Sigma}_{11}\right)\right)^{1/2}} \cdot \exp\left(-\frac{1}{2}\left(x_1-\tilde{\mu}_1\right)^T\left(\tilde{\Sigma}_{11}\right)^{-1}\left(x_1-\tilde{\mu}_1\right)\right) \tag{8}$$

where

$$\tilde{\mu}_1 = \mu_1 + \Sigma_{12}{\Sigma_{22}}^{-1}(x_2-\mu_2) \tag{9}$$

and

$$\tilde{\Sigma}_{11} = \Sigma_{11} - \Sigma_{12}{\Sigma_{22}}^{-1}\Sigma_{21}. \tag{10}$$

Equation (9) tells us that, when a sample x_2 in X_2 is given, the expectation of X_1 will change accordingly, and this change can be regarded as the correction of x_2 to X_1.

Based on this, we first compute the mean $\binom{\mu_k^{f^s}}{\mu_k^{f^q}}$ and the covariance matrix $\Sigma_k^f = \begin{pmatrix} \Sigma_k^{f^{ss}} & \Sigma_k^{f^{sq}} \\ \Sigma_k^{f^{qs}} & \Sigma_k^{f^{qq}} \end{pmatrix}$ of $F_k^f = \binom{f_k^{f^s}(\mathcal{R}_k^f)}{f_k^{f^q}}$, and the mean $\binom{\mu_k^{b^s}}{\mu_k^{b^q}}$ and the covariance matrix $\Sigma_k^b = \begin{pmatrix} \Sigma_k^{b^{ss}} & \Sigma_k^{b^{sq}} \\ \Sigma_k^{b^{qs}} & \Sigma_k^{b^{qq}} \end{pmatrix}$ of $F_k^b = \binom{f_k^{b^s}(\mathcal{R}_k^b)}{f_k^{b^q}}$. The conditional Gaussian probability density of $f_k^{f^s}(\mathcal{R}_k^f)$ given $f_k^{f^q}$ is as follows:

$$\begin{aligned} &\mathcal{D}_{f_k^{f^s}|f_k^{f^q}}\left(f_k^{f^s}\left(\mathcal{R}_k^f(\hat{i})\right)|f_k^{f^q}(\hat{i})\right) \\ &= \frac{1}{(2\pi)^{n_k^{f^s}/2}\left(\det\left(\widetilde{\Sigma_k^{f^{ss}}}\right)\right)^{1/2}} \\ &\quad\times \exp\left(-\frac{1}{2}\left(f_k^{f^s}\left(\mathcal{R}_k^f(\hat{i})\right)-\widetilde{\mu_k^{f^s}}\right)^T(\widetilde{\Sigma_k^{f^{ss}}})^{-1}\right. \\ &\quad\left.\times\left(f_k^{f^s}\left(\mathcal{R}_k^f(\hat{i})\right)-\widetilde{\mu_k^{f^s}}\right)\right) \end{aligned} \tag{11}$$

and the conditional Gaussian probability density of $f_k^{b^s}(\mathcal{R}_k^b)$ given $f_k^{b^q}$ is as follows:

$$\begin{aligned} &\mathcal{D}_{f_k^{b^s}|f_k^{b^q}}\left(f_k^{b^s}\left(\mathcal{R}_k^b(\hat{i})\right)|f_k^{b^q}(\hat{i})\right) \\ &= \frac{1}{(2\pi)^{n_k^{b^s}/2}\left(\det\left(\widetilde{\Sigma_k^{b^{ss}}}\right)\right)^{1/2}} \\ &\quad\times \exp\left(-\frac{1}{2}\left(f_k^{b^s}\left(\mathcal{R}_k^b(\hat{i})\right)-\widetilde{\mu_k^{b^s}}\right)^T\left(\widetilde{\Sigma_k^{b^{ss}}}\right)^{-1}\right. \\ &\quad\left.\times\left(f_k^{b^s}\left(\mathcal{R}_k^b(\hat{i})\right)-\widetilde{\mu_k^{b^s}}\right)\right). \end{aligned} \tag{12}$$

Each feature vector $f_k^{f^q}(\hat{i})$ in $f_k^{f^q}$ contributes to the correction of $\mu_k^{f^s}$, and each feature vector $f_k^{b^q}(\hat{i})$ in $f_k^{b^q}$ contributes to the correction of $\mu_k^{b^s}$

$$\begin{cases} \hat{\xi}_k^f(\hat{i}) = \mu_k^{f^s} + \Sigma_k^{f^{sq}}\Sigma_k^{f^{qq-1}}\left(f_k^{f^q}(\hat{i}) - \mu_k^{f^q}\right) \\ \hat{\xi}_k^b(\hat{i}) = \mu_k^{b^s} + \Sigma_k^{b^{sq}}\Sigma_k^{b^{qq-1}}\left(f_k^{b^q}(\hat{i}) - \mu_k^{b^q}\right) \end{cases} \tag{13}$$

where $\hat{\xi}_k^f(\hat{i})$ is the corrected $\mu_k^{f^s}$ given $f_k^{f^q}(\hat{i})$, and $\hat{\xi}_k^b(\hat{i})$ is the corrected $\mu_k^{b^s}$ given $f_k^{b^q}(\hat{i})$. When all the feature vectors

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in f_k^{fq} and f_k^{bq} are performed by (13), a new support feature vector set $\hat{\xi}_k^f$ for the kth category and a new support feature vector set $\hat{\xi}_k^f$ for the background are constructed

$$\begin{cases} \hat{\xi}_k^f = \left\{ \hat{\xi}_k^f(\hat{i}), \hat{i} = 1, \ldots, N_k^{fq} \right\} \\ \hat{\xi}_k^b = \left\{ \hat{\xi}_k^b(\hat{i}), \hat{i} = 1, \ldots, N_k^{bq} \right\}. \end{cases} \tag{14}$$

Then, f_k^{fs} and f_k^{bs} in (2)–(6) are replaced with $\hat{\xi}_k^f$ and $\hat{\xi}_k^b$. After these equations are performed, a new prediction P_k^{CGE} is generated.

2) Regularization: The premise of the enhancement process in Section III-D1 is that $F_k^f = \begin{pmatrix} f_k^{fs}(\mathcal{R}_k^f) \\ f_k^{fq} \end{pmatrix}$ and $F_k^b = \begin{pmatrix} f_k^{bs}(\mathcal{R}_k^b) \\ f_k^{bq} \end{pmatrix}$ follow the Gaussian distribution. The feature extractor cannot meet this premise. Here, we add a regularization term to meet this Gaussian distribution requirement.

The probability density of $F_k^f = \begin{pmatrix} f_k^{fs}(\mathcal{R}_k^f) \\ f_k^{fq} \end{pmatrix}$ is as follows:

$$\begin{aligned} &\mathcal{D}_{F_k^f}\begin{pmatrix} f_k^{fs}\left(\mathcal{R}_k^f(\hat{i})\right) \\ f_k^{fq}(\hat{i}) \end{pmatrix} \\ &= \frac{1}{(2\pi)^{C^{fc}}\left(\det\left(\Sigma_k^f\right)\right)^{1/2}} \\ &\quad \times \exp\left(-\frac{1}{2}\begin{pmatrix} f_k^{fs}\left(\mathcal{R}_k^f(\hat{i})\right) - \mu_k^{fs} \\ f_k^{fq}(\hat{i}) - \mu_k^{fq} \end{pmatrix}^T \left(\Sigma_k^f\right)^{-1} \right. \\ &\qquad \left. \times \begin{pmatrix} f_k^{fs}\left(\mathcal{R}_k^f(\hat{i})\right) - \mu_k^{fs} \\ f_k^{fq}(\hat{i}) - \mu_k^{fq} \end{pmatrix}\right) \end{aligned} \tag{15}$$

and the probability density of $F_k^b = \begin{pmatrix} f_k^{bs}(\mathcal{R}_k^b) \\ f_k^{bq} \end{pmatrix}$ is the following:

$$\begin{aligned} &\mathcal{D}_{F_k^b}\begin{pmatrix} f_k^{bs}\left(\mathcal{R}_k^b(\hat{i})\right) \\ f_k^{bq}(\hat{i}) \end{pmatrix} \\ &= \frac{1}{(2\pi)^{C^{fc}}\left(\det\left(\Sigma_k^b\right)\right)^{1/2}} \\ &\quad \times \exp\left(-\frac{1}{2}\begin{pmatrix} f_k^{bs}\left(\mathcal{R}_k^b(\hat{i})\right) - \mu_k^{bs} \\ f_k^{bq}(\hat{i}) - \mu_k^{bq} \end{pmatrix}^T \left(\Sigma_k^b\right)^{-1} \right. \\ &\qquad \left. \times \begin{pmatrix} f_k^{bs}\left(\mathcal{R}_k^b(\hat{i})\right) - \mu_k^{bs} \\ f_k^{bq}(\hat{i}) - \mu_k^{bq} \end{pmatrix}\right). \end{aligned} \tag{16}$$

Similar to maximum likelihood estimation, given a set of samples of F_k^f, assuming they are independent of each other, we need to maximize the joint probability density of this set of samples under the Gaussian distribution assumption. At this time, we optimize the parameters θ^{fg} of the feature generator because these parameters directly determine the attributes of F_k^f. For F_k^b, the process is similar. The form of the likelihood is the following:

$$\begin{cases} \mathcal{LF}_k^f = \prod_{\hat{i}=1}^{N_k^{fq}} \left(\mathcal{D}_{F_k^f}\begin{pmatrix} f_k^{fs}\left(\mathcal{R}_k^f(\hat{i})\right) \\ f_k^{fq}(\hat{i}) \end{pmatrix} \right) \\ \mathcal{LF}_k^b = \prod_{\hat{i}=1}^{N_k^{bq}} \left(\mathcal{D}_{F_k^b}\begin{pmatrix} f_k^{bs}\left(\mathcal{R}_k^b(\hat{i})\right) \\ f_k^{bq}(\hat{i}) \end{pmatrix} \right). \end{cases} \tag{17}$$

Taking the negative logarithm of (17), we transform the maximum likelihood into a loss minimization problem

$$\begin{cases} \mathcal{L}_k^f = -\log \mathcal{LF}_k^f \\ \mathcal{L}_k^b = -\log \mathcal{LF}_k^b \end{cases} \tag{18}$$

$$\mathcal{L}_k^R = \mathcal{L}_k^f + \mathcal{L}_k^b. \tag{19}$$

The negative logarithm not only transforms the maximization problem into a minimization problem, but also avoids the out-of-bounds problem that may occur when calculating the determinant of a large matrix.

3) Degenerated Case: In a special case, Σ_k^f or Σ_k^b has reduced rank, which forms singular Gaussian distributions. As a result, $\hat{\xi}_k^f(\hat{i})$ or $\hat{\xi}_k^b(\hat{i})$ in (13) cannot be computed, so the optimization using (18) is invalid. Based on the conclusion of [60], given a centered vector $X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$ that follows a singular Gaussian with the degenerate covariance matrix $\begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$, the optimal (Bayesian) estimate of the vector X_1 is in the form:

$$\hat{X}_1 = \Sigma_{12}\Sigma_{22}^{\mathcal{MP}} X_2 \tag{20}$$

where $\Sigma_{22}^{\mathcal{MP}}$ is the Moore–Penrose inverse of Σ_{22}

$$\Sigma_{22}^{\mathcal{MP}} = \Sigma_C^T\left(\Sigma_C\Sigma_C^T\right)^{-1}\left(\Sigma_B^T\Sigma_B\right)^{-1}\Sigma_B^T \tag{21}$$

where $\Sigma_{22} = \Sigma_B\Sigma_C$ is the full rank decomposition of Σ_{22}.

In terms of the enhancement, if Σ_k^{fqq} and Σ_k^{bqq} are degenerated, according to (20) and (21) and considering the centralization process of (20), $\hat{\xi}_k^f$ and $\hat{\xi}_k^b$ will be computed as follows:

$$\begin{cases} \hat{\xi}_k^f(\hat{i}) = \mu_k^{fs} + \Sigma_k^{fsq}\Sigma_k^{fqq\mathcal{MP}}\left(f_k^{fq}(\hat{i}) - \mu_k^{fq}\right) \\ \hat{\xi}_k^b(\hat{i}) = \mu_k^{bs} + \Sigma_k^{bsq}\Sigma_k^{bqq\mathcal{MP}}\left(f_k^{bq}(\hat{i}) - \mu_k^{bq}\right) \end{cases} \tag{22}$$

where $\Sigma_k^{fqq\mathcal{MP}}$ is the Moore–Penrose inverse of Σ_k^{fqq}. Then, a new support feature vector set $\hat{\xi}_k^f$ for the kth category and a new support feature vector set $\hat{\xi}_k^b$ for the corresponding background are constructed

$$\begin{cases} \hat{\xi}_k^f = \left\{ \hat{\xi}_k^f(\hat{i}), \hat{i} = 1, \ldots, N_k^{fq} \right\} \\ \hat{\xi}_k^b = \left\{ \hat{\xi}_k^b(\hat{i}), \hat{i} = 1, \ldots, N_k^{bq} \right\}. \end{cases} \tag{23}$$

In terms of the regularization, the degenerated Σ_k^{fqq} and Σ_k^{bqq} poses challenges regarding the computation of the probability density. Here, we use a way to perform regularization that can be intuitively explained. Given $F_k^f = \begin{pmatrix} f_k^{fs}(\mathcal{R}_k^f) \\ f_k^{fq} \end{pmatrix}$ and $F_k^b = \begin{pmatrix} f_k^{bs}(\mathcal{R}_k^b) \\ f_k^{bq} \end{pmatrix}$, the covariance matrices M_k^f and M_k^b are computed as follows:

$$\begin{cases} M_k^f = \left(F_k^f - \mu_k^f\right) \cdot \left(F_k^f - \mu_k^f\right)^T \\ M_k^b = \left(F_k^b - \mu_k^b\right) \cdot \left(F_k^b - \mu_k^b\right)^T \end{cases} \tag{24}$$

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where $\mu_k^f = \begin{pmatrix} \mu_k^{f^s} \\ \mu_k^{f^q} \end{pmatrix}$ and $\mu_k^b = \begin{pmatrix} \mu_k^{b^s} \\ \mu_k^{b^q} \end{pmatrix}$. The losses based on M_k^f and M_k^b can be computed as follows:

$$\begin{cases} \mathcal{L}_k^{f^t} = \dfrac{1}{D}\mathrm{Tr}\left(M_k^f\right) \\ \mathcal{L}_k^{b^t} = \dfrac{1}{D}\mathrm{Tr}\left(M_k^b\right) \end{cases} \tag{25}$$

where D is the dimension of M_k^f and M_k^b, and $\mathrm{Tr}(\cdot)$ is the operator used to determine the trace of a matrix. Then, to make the support features and query features closer, we calculate the following loss:

$$\begin{cases} \mathcal{L}_k^{f^m} = \left(\mu_k^{f^s} - \mu_k^{f^q}\right)^T \cdot \left(\mu_k^{f^s} - \mu_k^{f^q}\right) \\ \mathcal{L}_k^{b^m} = \left(\mu_k^{b^s} - \mu_k^{b^q}\right)^T \cdot \left(\mu_k^{b^s} - \mu_k^{b^q}\right). \end{cases} \tag{26}$$

Finally, the losses for the degenerated case are formulated

$$\begin{cases} \mathcal{L}_k^f = \mathcal{L}_k^{f^t} + \mathcal{L}_k^{f^m} \\ \mathcal{L}_k^b = \mathcal{L}_k^{b^t} + \mathcal{L}_k^{b^m} \end{cases} \tag{27}$$

$$\mathcal{L}_k^R = \mathcal{L}_k^f + \mathcal{L}_k^b. \tag{28}$$

E. Combined Prediction

P_k^{DCM} and P_k^{CGE} are first combined

$$P_k^{\mathrm{en}} = \left(\mathrm{Softmax}\left(P_k^{\mathrm{DCM}}\right) + \mathrm{Softmax}\left(P_k^{\mathrm{CGE}}\right)\right)/2.0. \tag{29}$$

Subsequently, a enhanced query feature $\hat{f}^q$ is obtained using f^q and P_k^{en}

$$\hat{f}^q = f^q * P_k^{\mathrm{en}}[1] \tag{30}$$

where $*$ is the element-by-element product. Then, $\hat{f}^q$ is put into a prediction layer that is as same as that of the FCN-based semantic segmentation networks [61] to produce the final prediction P_k^F for the kth category.

F. Optimization

To optimize the parameters θ_{fg} in the feature generator, three cross-entropy losses $\mathcal{L}_k^{\mathrm{DCM}}$, $\mathcal{L}_k^{\mathrm{CGE}}$, and $\mathcal{L}_k^F$ are produced and the regularization term $\mathcal{L}_k^R$ are employed

$$\begin{cases} \mathcal{L}_k^{\mathrm{DCM}} = \mathrm{CE}\left(P_k^{\mathrm{DCM}}, Y^s[M_k]\right) \\ \mathcal{L}_k^{\mathrm{CGE}} = \mathrm{CE}\left(P_k^{\mathrm{CGE}}, Y^s[M_k]\right) \\ \mathcal{L}_k^F = \mathrm{CE}\left(P_k^F, Y^s[M_k]\right) \end{cases} \tag{31}$$

where $\mathrm{CE}(\cdot)$ computes the cross-entropy between the predictions and ground-truth. The cross-entropy is well-known, and we do not present it here. M_k is the mask of the kth category in Y^s. Then, the optimization process is as follows:

$$\min_{\theta_{\mathrm{fg}}}\left(\mathcal{L}_k^{\mathrm{DCM}} + \mathcal{L}_k^{\mathrm{CGE}} + \mathcal{L}_k^F + \mathcal{L}_k^R\right). \tag{32}$$

G. Land Cover Classification

After the CGE-DCM has been trained, CGE-DCM can be directly applied to generate land cover maps. As shown in Fig. 1, an initial prediction map $P \in \mathbb{R}^{K\times H\times W}$ filled with zero is first created. Then, each run of CGE-DCM generates a prediction P_k^F for a category (denoted by k). After P_k^F is upsampled and transformed into the image plane $\mathbb{R}^{2\times H\times W}$, P is updated as follows:

$$\begin{cases} P = P(1,:,:) + P_1^F(1,:,:) \\ \quad\vdots \\ P = P(k,:,:) + P_k^F(1,:,:) \\ \quad\vdots \\ P = P(K,:,:) + P_K^F(1,:,:). \end{cases} \tag{33}$$

According to (33), in each update, the probability of each pixel belonging to a certain category is recorded by P. When K times of updates are completed, the probability of each pixel belonging to all K categories is recorded. Finally, the land cover map LC is obtained as follows:

$$LC = \left\{ \arg\max_{k\in\{1,\ldots,K\}} P(:,h,w),\ h = 1,\ldots,H,\ w = 1,\ldots,W \right\}. \tag{34}$$

IV. Experiments

We first introduce the datasets including GID [9] and WHU-OHS [64] for testing CGE-DCM. Next, the experimental settings are presented, and then, the quantitative and visual results are shown. Finally, an ablation study is performed to present the impact of each component on CGE-DCM.

A. Datasets

1) GID: The GID is constructed based on the Gaofen-2 satellite images. Gaofen-2 is the second satellite of the high-definition Earth observation system (HDEOS) promoted by the China National Space Administration (CNSA). Detailed satellite parameters can be found in [9]. The GID contains 150 high-quality Gaofen-2 images acquired from more than 60 different cities in China. Each image contains 6800 × 7200 pixels that cover a geographic area of 506 km^2. The spatial resolution is about 3.1–3.3 m. The GID uses the Chinese Land Use Classification Criteria (GB/T21010-2017) to determine a hierarchical category system. The hierarchical category system forms a large-scale classification set and a fine-scale classification set. In the former set, five large-scale categories as shown in the first column of Table I are annotated. In the latter set, 15 fine-scale categories as shown in the last column of Table I are annotated. The large-scale classification set is used for training, and the fine-scale classification set is used for testing. In this setting, when CGE-DCM is training, CGE-DCM cannot see the fine-scale categories. We crop the original images containing fine-scale categories into 1000 nonoverlapping images, each of which contains 680 × 720 pixels.

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TABLE II
HIERARCHICAL CATEGORY SYSTEMS OF WHU-OHS

Large-scale categories	Fine-scale categories
Farmland	Paddy field Dry farm
Forest	Woodland Shrubbery Sparse woodland Other forest land
Grassland	High-covered grassland Medium-covered grassland Low-covered grassland
Water body	River/canal Lake Reservoir/pond Beach land Shoal
Built-up land	Urban built-up Rural settlement Other construction land
Unused land	Sand Gobi Saline/alkali soil Marshland Bare land bare rock
Ocean	Marine area

2) WHU-OHS: The WHU-OHS dataset is composed of 7795 Orbita hyperspectral satellite (OHS) images acquired from more than 40 different locations in China. Each image in WHU-OHS contains 512 × 512 pixels with a spatial resolution of 10 m. The OHS images are collected by the hyperspectral constellation composed of ten Orbita hyperspectral micro-nano satellites which are managed by Zhuhai Orbita Aerospace, China. The hyperspectral constellation can achieve global observation within two days. Detailed satellite parameters can be found in [64]. Like GID, WHU-OHS also uses the Chinese Land Use Classification Criteria (GB/T21010-2017) to determine the category system. Twenty-four fine-scale categories as shown in Table II are contained in the WHU-OHS. Here, 75% images are used for training, and 25% images are used for testing. To validate the performance of CGE-DCM in the cross-category LCC task, we combine the fine-scale categories into large-scale categories in the training set and keep the testing set unchanged. The large-scale categories include *farmland, forest, grassland, water body, built-up land, unused land,* and *ocean* as shown in Table II. Because the ocean category only contains one subcategory (marine area), here considering the cross-category setting, we ignore the ocean in the large-scale categories and marine area in the fine-scale categories.

B. Experimental Settings

1) Evaluation Metrics: We employ the intersection-over-union (IoU) score of each category and the mean IoU (mIoU) score of all existing categories as the basis for our accuracy evaluation. As discussed in Section I, we allow for only one annotated example of each fine-scale category to the model when mapping fine-scale land cover. The different selections of the annotated example lead to different results. To make the evaluation more objective, we repeatedly test the model 20× and obtain the LCC results by using a completely different set of annotated examples for each test. The result of each test generates a set of IoU scores $\{\text{IoU}_k^i,\ k = 1, \ldots, K\}$ and a mIoU score mIoU^i, where K is the total number of categories and $i = 1, \ldots, 20$ indicates the ith test.

To evaluate the overall accuracy, we calculate the average mIoU (mIoU^{ave}) score based on $\{\text{mIoU}^i,\ i = 1, \ldots, 20\}$ and take the minimum and maximum values ($\text{mIoU}^{\min}$ and $\text{mIoU}^{\max}$) in $\{\text{mIoU}^i,\ i = 1, \ldots, 20\}$ to evaluate the worst and best accuracy. At the same time, we calculate the standard deviation Dev^{mIoU} of mIoU scores $\{\text{mIoU}^i,\ i = 1, \ldots, 20\}$ to measure the stability of the overall performance of the method. Concretely,

$$\begin{cases} \text{mIoU}^{\text{ave}} = \dfrac{1}{20}\displaystyle\sum_i^{20} \text{mIoU}^i \\ \text{mIoU}^{\min} = \min\limits_i \left\{\text{mIoU}^i,\ i = 1, \ldots, 20\right\} \\ \text{mIoU}^{\max} = \max\limits_i \left\{\text{mIoU}^i,\ i = 1, \ldots, 20\right\} \end{cases} \tag{35}$$

$$\text{Dev}^{\text{mIoU}} = \sqrt{\frac{1}{20}\sum_i^{20}\left(\text{mIoU}^i - \text{mIoU}^{\text{ave}}\right)^2}. \tag{36}$$

There are many fine-scale categories of land cover. For example, the WHU-OHS dataset contains 24 fine-scale categories. The limited space makes it difficult to list the IoU scores of all categories. Even if they are listed, it is still difficult to find patterns behind the data when there are too many values. Therefore, this article defines IoU-based variances. First, the variance Var^i of the IoU scores $\{\text{IoU}_k^i,\ k = 1, \ldots, K\}$ of each test is computed. Then, the minimum and maximum variances ($\text{Var}^{\min}$ and $\text{Var}^{\max}$) in $\{\text{Var}^i,\ i = 1, \ldots, 20\}$ are given, and the average variance Var^{ave} of $\{\text{Var}^i,\ i = 1, \ldots, 20\}$ is calculated to reflect the stability of the model's performance on different categories. Concretely, these IoU-based variances are defined as follows:

$$\begin{cases} \text{Var}^{\min} = \min\limits_i \left\{\text{Var}^i,\ i = 1, \ldots, 20\right\} \\ \text{Var}^{\max} = \max\limits_i \left\{\text{Var}^i,\ i = 1, \ldots, 20\right\} \\ \text{Var}^{\text{ave}} = \dfrac{1}{20}\displaystyle\sum_i^{20} \text{Var}^i. \end{cases} \tag{37}$$

2) Implementation Details: To train CGE-DCM, we use a stochastic gradient descent (SGD) optimizer. The initial learning rate is set to 1×10^{-4}, the momentum is set to 0.9, and the weight decay is set to 5×10^{-5}. The training process is iterated 100 000×, and the total number of iterations is divided into three stages. The learning rate of each stage is one-tenth of that of the previous stage. We use ResNet-50 [58] as the feature generator to produce f^s and f^q, as shown in Fig. 1. In both the GID and WHU-OHS datasets, we use the large-scale categories for training, and we use

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TABLE III
COMPARISON METHODS

Methods	Modules	Losses	Description
PANet [44]	(1) Support-to-query segmentation and (2) query-to-support segmentation	(1) Loss for support-to query segmentation and (2) loss for query-to-support segmentation	PANet is one of the models that established a basic framework for few-shot segmentation. It learns class-specific prototype representations and matches each pixel using cosine distances.
PPNet [53]	(1) Embedding network; (2) prototype generation network, including a part generation module and a part refinement module; and (3) a part-aware mask generation network	(1) Loss for support-to-query segmentation, (2) loss for query-to-support segmentation, and (3) loss for the global semantic class space	PPNet decomposes the holistic class representation into a set of part-aware prototypes and leverages unlabeled data to enrich these prototypes.
PMMs [21]	(1) Prototype mixture models, (2) duplex strategy for semantic learning, and (3) residual prototype mixture models	Cross-entropy loss at the query branch	PMMs are prototype mixture models that correlate diverse image regions with multiple prototypes to enforce prototype-based semantic representation.
PFENet [52]	(1) Prior generation module, (2) feature enrichment module (FEM), and (3) prior guided feature enrichment network	(1) A set of losses for the intermediate supervision produced by FEM, and (2) loss for the final prediction	PFENet uses the ImageNet [62] pretrained high-level feature to produce prior mask. FEM is then proposed to adaptively enrich query features with support features and prior masks.
SSP [22]	(1) Self-support prototype generation, (2) adaptive self-support background prototype generation, and (3) self-support matching	(1) Loss for the distance maps, (2) query self-support loss, and (3) support self-matching loss	SSP designs a novel self-support matching strategy that produces both the self-support foreground and background prototypes.
HPA [11]	(1) Training-free base prototype acquisition, (2) prototype activation module (PAM), and (3) cross-referenced decoder (CRD)	(1) Segmentation loss and (2) activation loss	HPA, which combines the training-free base prototype acquisition, PAM, and CRD, investigates the transferability and scalability of few-shot segmentation models.
MGCL [56]	(1) Mask-guided feature enhancement, (2) foreground-background correlation, (3) mask-guided correlation decoder, and (4) the frozen segment anything model [63]	The binary cross-entropy loss for one-shot learning	MGCL enhances the contrast between foreground and background and learns how to segment complete objects by focusing on the consistency within the mask.
CGE-DCM	(1) DCM and (2) CGE	L_k^{DCM}, L_k^{GCE}, L_k^F and L_k^R	CGE-DCM introduces DCM and CGE to deal with complex scenes in remote sensing tasks and differences between context distributions in different category systems.

the fine-scale categories for testing. For more details, in the GID, the large-scale classification set is used for training, and the fine-scale land-cover classification set is used for testing. In WHU-OHS, the training set with combined large-scale categories is used for training, and the testing set with fine-scale categories is used for testing.

3) Comparison Methods: For comparison, we have selected seven representative few-shot segmentation methods, including PANet [44], PPNet [53], PMMs [21], PFENet [52], SSP [22], HPA [11], and MGCL [56]. The detailed information on these methods is shown in Table III. In order to ensure the fairness of the comparison of experimental results, for all methods, only one annotated example of each fine-scale category is provided as supporting information for all the query images during the testing process.

C. Results in GID

The results in GID are divided into two parts: quantitative and visual comparisons. In these comparison results, we can find the advantages of CGE-DCM.

1) Quantitative Comparison in GID: The quantitative comparison in the GID is shown in Table IV. We compare CGE-DCM with the existing methods presented in Section IV-B3. As shown in Table IV, CGE-DCM obtains the highest mIoU^{ave}, mIoU^{min}, and mIoU^{max} scores. This shows that, in the random tests, CGE-DCM achieves the highest average accuracy, best accuracy, and worst accuracy compared with the other methods. Compared with MGCL, which is the second most accurate method, CGE-DCM has an average accuracy that is 3.64% higher, a worst accuracy of 5.01% higher, and a best accuracy of 4.08% higher. Simply having a higher accuracy sometimes cannot fully reflect the stability of the method. Therefore, we calculate the standard deviation Dev^{mIoU} of mIoU scores, as shown in (36), and the IoU-based variances Var^{min}, Var^{ave}, and Var^{max} as shown in (37). From Table IV, CGE-DCM obtains the lowest Dev^{mIoU} and lowest IoU-based variances. This shows that our method is more stable than existing methods in terms of both overall accuracy and category-level accuracy.

2) Visual Comparison in GID: As shown in Table IV, SSP, HPA, and MGCL obtain a higher accuracy than the other existing methods; we only present the visual results of SSP, HPA, MGCL, and our CGE-DCM in Fig. 2. Four representable areas are selected to show the visual comparison. Because

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TABLE IV
QUANTITATIVE RESULTS IN GID

Methods	mIoU based scores			mIoU deviation	IoU-based variances		
	$mIoU^{ave}(\%)$	$mIoU^{min}(\%)$	$mIoU^{max}(\%)$	$Dev^{mIoU}(‰)$	$Var^{ave}(\%)$	$Var^{min}(\%)$	$Var^{max}(\%)$
PANet [44]	46.73	45.15	49.11	0.41	6.52	6.05	6.98
PPNet [53]	47.24	46.63	49.41	0.51	6.55	6.11	6.99
PMMs [21]	47.82	46.31	51.17	0.48	6.96	6.02	7.41
PFENet [52]	49.52	47.13	51.49	0.47	6.39	5.40	7.04
SSP [22]	50.05	48.17	52.40	0.48	6.38	5.57	6.90
HPA [11]	50.33	48.27	52.51	0.46	6.59	5.92	7.08
MGCL [56]	50.83	48.52	53.04	0.45	6.39	5.36	7.02
CGE-DCM(Ours)	**54.47**	**53.53**	**57.12**	**0.21**	**6.26**	**5.35**	**6.67**

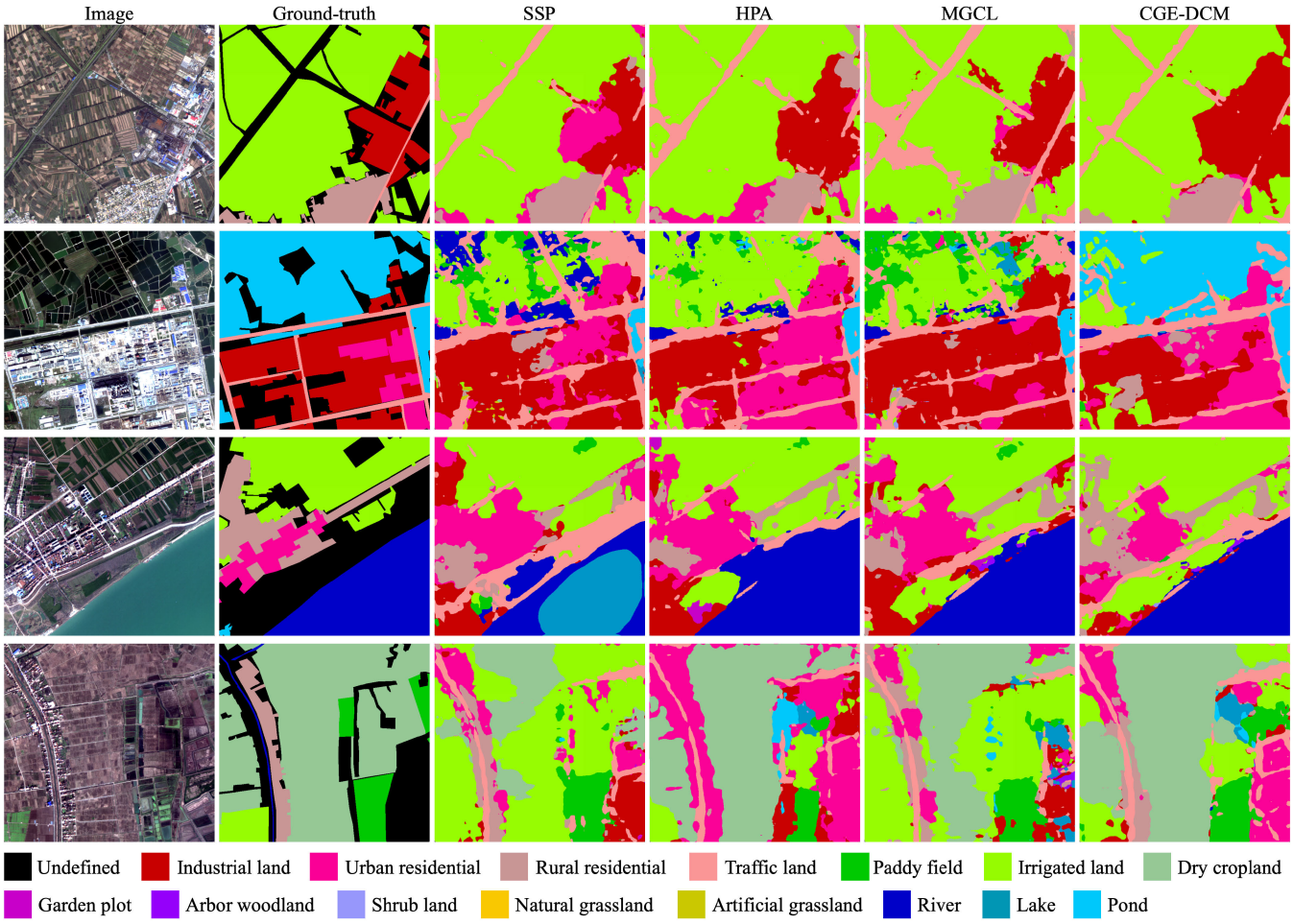


Fig. 2. Visual results of the cross-category LCC in GID.

the land cover categories of the GID are finely divided and the spectral and contextual characteristics of many categories are similar, there are numerous misjudgments in the results of each method. As shown in the last column of Fig. 2, the results of CGE-DCM are still significantly better than other methods. For example, in the first area, SSP misidentifies some industrial land as urban residential areas, HPA misidentifies some rural residential areas as urban residential areas, and MGCL misidentifies a part of irrigated land as industrial land. In the third area, SSP misidentifies the river as lakes, and the boundaries of the river identified by HPA and MGCL are inaccurate, but CGE-DCM does not have these problems.

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TABLE V
QUANTITATIVE RESULTS IN WHU-OHS

Methods	mIoU based scores			mIoU deviation	IoU-based variances		
	$mIoU^{ave}(\%)$	$mIoU^{min}(\%)$	$mIoU^{max}(\%)$	$Dev^{mIoU}(‰)$	$Var^{ave}(\%)$	$Var^{min}(\%)$	$Var^{max}(\%)$
PANet [44]	24.91	21.93	27.02	0.91	5.06	4.09	6.05
PPNet [53]	26.72	23.26	28.85	0.86	5.27	4.79	6.26
PMMs [21]	27.53	24.11	31.04	0.92	5.11	4.54	6.01
PFENet [52]	29.28	25.15	32.70	0.93	5.10	4.51	6.03
SSP [22]	30.13	26.05	33.94	0.95	5.13	4.56	6.24
HPA [11]	31.51	27.42	33.90	0.79	5.06	4.49	6.11
MGCL [56]	31.62	27.57	33.88	0.83	5.09	4.50	6.04
CGE-DCM(Ours)	**35.16**	**30.48**	**38.13**	**0.47**	**4.97**	**4.05**	**5.97**

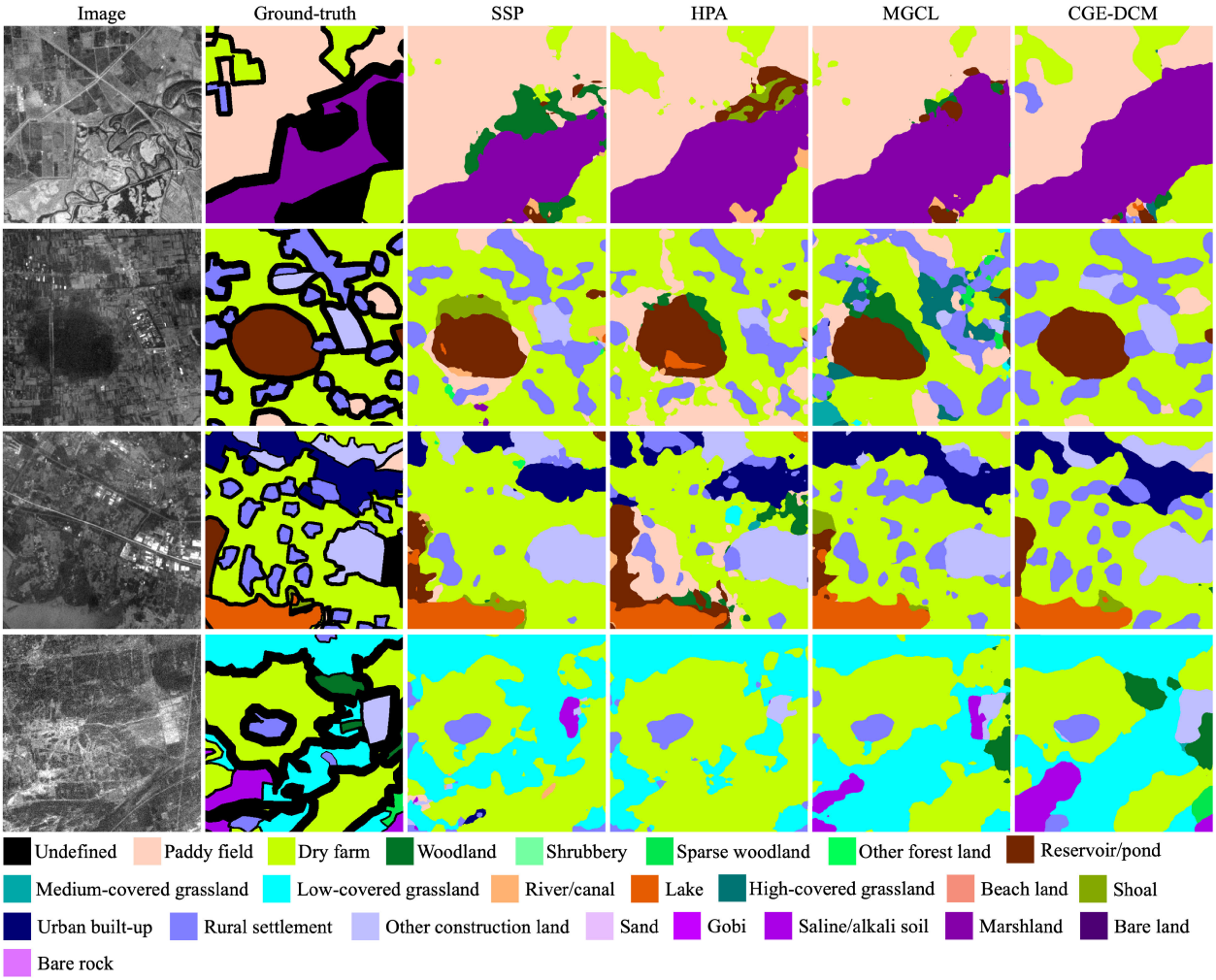


Fig. 3. Visual results of the cross-category LCC in WHU-OHS.

D. Results in WHU-OHS

The structure of this section is the same as that of Section IV-C, which is composed of quantitative and visual comparisons.

1) Quantitative Comparison in WHU-OHS: Table V shows the quantitative comparisons in WHU-OHS. Our CGE-DCM obtains the highest $mIoU^{ave}$, $mIoU^{min}$, and $mIoU^{max}$ scores among all the methods. The $mIoU^{ave}$ and $mIoU^{min}$ scores of

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TABLE VI
QUANTITATIVE ABLATION STUDY

Methods	mIoU based scores			mIoU deviation	IoU-based variances		
	$mIoU^{ave}$(%)	$mIoU^{min}$(%)	$mIoU^{max}$(%)	Dev^{mIoU}(‰)	Var^{ave}(%)	Var^{min}(%)	Var^{max}(%)
Baseline	39.13	36.22	42.64	0.35	6.57	6.02	7.08
CGE-DCM − DCM	50.17	46.74	53.13	0.31	6.73	6.03	7.15
CGE-DCM − FcmG	52.23	49.74	54.41	0.27	6.54	5.95	7.14
CGE-DCM − DenseM	51.33	47.50	53.92	0.28	6.65	5.97	7.16
CGE-DCM − CGE	52.06	50.91	55.17	0.48	7.25	6.38	7.97
CGE-DCM − Reg	53.31	50.26	56.07	0.44	7.03	6.11	7.69
CGE-DCM − DegC	53.35	50.32	56.10	0.42	7.01	6.20	7.66
CGE-DCM	**54.47**	**53.53**	**57.12**	**0.21**	**6.26**	**5.35**	**6.67**

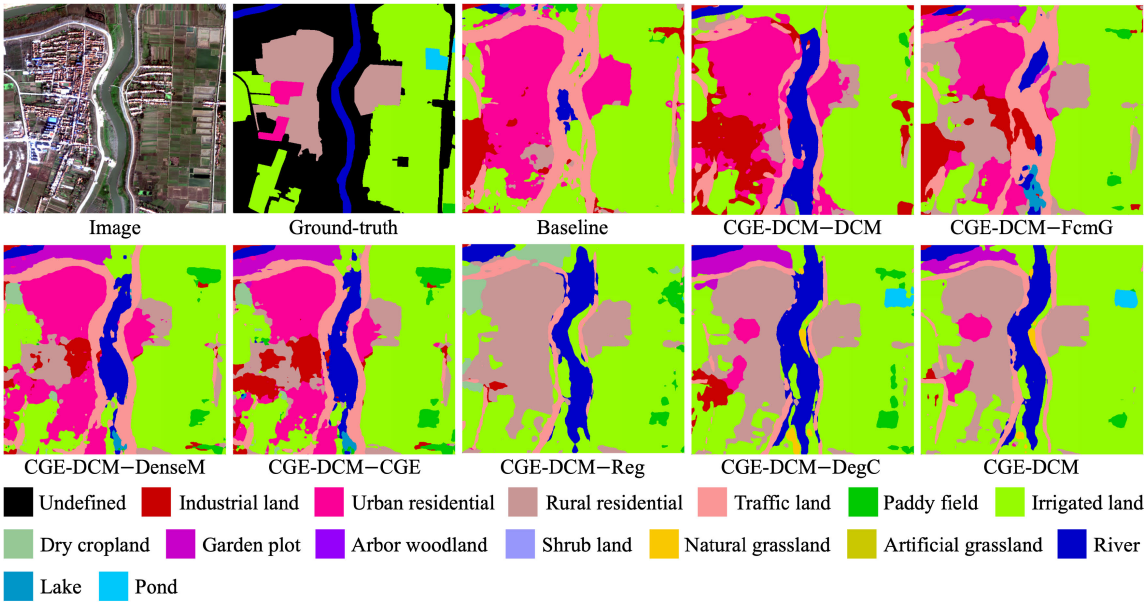


Fig. 4. Visual results of the ablation study.

CGE-DCM are 3.54% and 2.91% higher than that of MGCL which is ranked second. The $mIoU^{max}$ score of CGE-DCM is 4.19% higher than that of SSP. This demonstrates that CGE-DCM can obtain a higher overall accuracy than the existing methods. Furthermore, the standard deviation Dev^{mIoU} of mIoU scores of CGE-DCM is the lowest among all the methods, which shows that the performance of CGE-DCM is more stable regarding the overall accuracy. In terms of the category-level metrics, CGE-DCM obtains the lowest IoU-based variances (Var^{min}, Var^{ave}, and Var^{max}). This indicates that the complex spectral and contextual information of a specific category has less impact on CGE-DCM; thus, CGE-DCM achieves smaller variances in accuracy at the category level.

2) Visual Comparison in WHU-OHS: Fig. 3 shows the visual comparisons between CGE-DCM and the existing methods, including SSP, HPA, and MGCL. Because SSP, HPA, and MGCL obtain a higher accuracy than the other existing methods, we only show the visual results of them. WHU-OHS contains many categories, and the spectral confusion between different classes of objects is more obvious. Therefore, there are many misjudgments in the LCC results of all methods. For example, the boundaries of the marshland in the first area, the reservoir in the second area, and other construction land in the third and fourth areas recognized by these methods are all different from that of the ground-truth. However, in terms of the overall effect, the classification results of CGE-DCM are better. For example, in the second area, the identified reservoir by CGE-DCM is complete and not confused with other categories, while the recognition results of the other methods all have misjudgments regarding the reservoir. In the third area, all the small rural settlements are identified by CGE-DCM, while other methods miss most of the rural settlements. In the

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fourth area, although the boundaries of other construction land identified by CGE-DCM are inaccurate, the misjudgments of other construction land extracted by other methods are even more serious.

In sum, CGE-DCM obtains superior results than existing methods according to the quantitative and visual comparisons on both the GID and WHU-OHS datasets.

E. Ablation Study

This article conducts an ablation study of each module of CGE-DCM from both quantitative and visual perspectives. Table VI shows the quantitative ablation study. The baseline is the basic one-shot segmentation framework that is formulated by Sg-one [43]. The baseline is also the foundation of CGE-DCM. We first remove each module from CGE-DCM and then infer the impact of the module based on a decrease in accuracy and stability. As shown in Section III-C, the core of DCM includes two parts: FcmG and DenseM. We delete DCM, FcmG, and DenseM from CGE-DCM to form three simplified methods, which are denoted as CGE-DCM—DCM, CGE-DCM—FcmG, and CGE-DCM—DenseM, respectively. Another key component of CGE-DCM is the CGE. As shown in Section III-D, CGE includes conditional enhancement, regularization, and the degenerate case (DegC). Among these modules, conditional enhancement is indispensable; otherwise, the CGE cannot be formed. Therefore, conditional enhancement cannot be directly removed from CGE-DCM. By deleting CGE, regularization, and DegC, three simplified methods are formed, which are denoted as CGE-DCM—CGE, CGE-DCM—Regularization (Reg), and CGE-DCM—DegC, respectively.

As shown in Table VI, when DCM is removed from CGE-DCM, the accuracy decreases most significantly, but the standard deviation of mIoU scores and IoU-based variances are not substantially affected. This shows the importance of DCM in improving the accuracy of CGE-DCM. In addition, the two core parts of DCM, namely, FcmG and DenseM, have similar functions as DCM. In contrast, CGE does not have as much of an impact on the accuracy as DCM, but CGE has a significant impact on the stability of CGE-DCM. When CGE or a part of CGE is removed, the standard deviation of mIoU scores and IoU-based variances significantly increase. Visual ablation studies as shown in Fig. 4 can also derive the same rules. It is worth noting that, when CGE or a part of CGE is removed, more noises appear in the classification results, which further illustrates the positive role of CGE in the stability of CGE-DCM.

V. Conclusion

This article proposes a cross-category LCC method: CGE-DCM. CGE-DCM uses samples in a large-scale category system for training, and the model is then equipped with the ability to handle more refined category systems. CGE-DCM consists of two main modules: DCM and CGE. DCM constructs feature-correlation matrices and a dense matching process to fully uncover classification clues. CGE is used to enhance the prototype's ability to express unknown contexts. To evaluate the classification results, this article defines a set of evaluation metrics based on IoU scores, making the accuracy assessment more reliable.

Both the GID and WHU-OHS datasets are employed to test CGE-DCM and enable comparisons with existing methods on the cross-category LCC task. The quantitative and visual results show that CGE-DCM achieves higher accuracy and more stable classification results than existing methods. Furthermore, an ablation study demonstrates the positive influence of each of the modules in CGE-DCM. Specifically, DCM plays a vital role when it comes to improving classification accuracy, while CGE contributes to the stability of classification performance.

Although CGE-DCM has achieved better results than existing methods based on a comparison using the GID and WHU-OHS datasets, there is still room for improvement. For example, there are many ways to handle a DegC in CGE. In the future, we intend to study this issue further and develop a more concise and effective method.

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